Sol 4 (a) $\quad f(x) = 2e^{-2x} \quad x > 0$

$$F(x) = 1 - e^{-2x},$$

jt cdf $P(X_{(1)} \leq x,\ X_{(3)} \leq y)$ where $0 < x < y$

$$= P(X_{(3)} \leq y) - P(X_{(1)} > x,\ X_{(3)} \leq y)$$

$$= F(y)^3 - [F(y) - F(x)]^3$$

$$= (1 - e^{-2x})^3 - (e^{-2x} - e^{-2y})^3 \quad (02)$$

jt pdf $\quad P(X_{(1)} \leq x,\ X_{(3)} \leq y)$, $0 < x < y$

$$f_{(1),(3)}(x,y) = \frac{\partial^2 P(X_{(1)} \leq x,\ X_{(3)} \leq y)}{\partial x \partial y}$$

$$= \frac{\partial^2}{\partial x \partial y}\left[(1 - e^{-2x})^3 - (e^{-2x} - e^{-2y})^3\right]$$

$$= \frac{\partial}{\partial y}\left[3(1 - e^{-2x})^2 \times (+2) \times e^{-2x} - 3(e^{-2x} - e^{-2y})^2 (-2) e^{-2x}\right] \quad (02)$$

$$= \frac{\partial}{\partial y}\left[6e^{-2x}(e^{-2x} - e^{-2y})^2\right]$$

$$= 6e^{-2x} \times 2(e^{-2x} - e^{-2y})\, 2e^{-2y}$$

$$= 24e^{-2(x+y)}(e^{-2x} - e^{-2y}) \quad \text{where } 0 < x < y$$

6 (a) Given

Sol 5

X: diameter of any randomly selected ball bearing

P = P(a ball bearing will be rejected due to falling outside of the specification)

$= P(X < 2.99 \text{ or } X > 3.01)$

$= 1 - P(2.99 \leq X \leq 3.01)$

$= 1 - P\left(\frac{2.99-3}{0.005} < \frac{X-3}{0.005} < \frac{3.01-3}{0.005}\right)$

$= 1 - P(-2 \leq Z \leq 2)$ $[Z \sim N(0,1)]$

$= 1 - [\Phi(2) - \Phi(-2)]$

$= 1 - (1 - 2 \times 0.0228)$

$= 1 - 0.9544$

$= 0.0456$

03

Y: number of defective ball bearings out of 10 randomly selected ball bearings

$P(Y \leq 1) = \binom{10}{0} p^0 q^{10} + \binom{10}{1} p^1 q^9$

$(q = 1 - p)$

$= 0.6270 + 0.2995$

$= 0.9266$

02